

# ISPilot API - Production Status Report

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**Date:** August 28, 2026

**Status:**  **LIVE IN PRODUCTION**

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## Executive Summary

ISPilot API is now fully operational on Google Cloud Run with proper **Workload Identity** authentication. The service successfully communicates with Vertex AI Reasoning Engines and responds to natural language queries about retail performance.

**Service URL:** <https://ispilot-api-46y2f3tyja-uc.a.run.app>

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## ☒ What's Working

### Core Functionality

- ☒ Chat endpoint responding correctly
- ☒ Vertex Reasoning Engine integration operational
- ☒ Session management (automatic creation)
- ☒ Natural language query processing
- ☒ Retail KPI analysis
- ☒ Store performance recommendations

### Authentication & Security

- ☒ Workload Identity (primary mechanism)
- ☒ OAuth Bearer token support
- ☒ API key fallback (local development)
- ☒ Cloud Run security: `--no-allow-unauthenticated`
- ☒ Service account isolation (`sa-tot-osa`)

### Infrastructure

- ☒ Docker multi-stage build (optimized)
- ☒ Cloud Run deployment automated
- ☒ Artifact Registry integration
- ☒ Cloud Logging configured
- ☒ Health checks passing
- ☒ Session storage (in-memory + Firestore fallback)

### Code Quality

- ☒ No hardcoded credentials
- ☒ Environment-based configuration

- ☒ Error handling implemented
- ☒ Logging infrastructure in place
- ☒ Request ID tracking

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## Recent Fixes (August 28, 2026)

### 1. Workload Identity Authentication

**Issue:** Cloud Run was running with hardcoded `GOOGLE_APPLICATION_CREDENTIALS` environment variable

**Solution:** Removed environment variable export, enabled pure Workload Identity

**Files Changed:** `deploy.sh`, `app/services/vertex_client.py`

**Result:** ☒ Service now authenticates cleanly via Workload Identity

### 2. Deployment Script Corrections

**Issue:** Environment variable formatting errors in `gcloud run deploy`

**Solution:** Fixed `--set-env-vars` to be comma-separated single string

**Impact:** Deployment now succeeds without gcloud argument errors

### 3. Secret Management

**Issue:** Referenced non-existent secret in deployment

**Solution:** Updated to correct secret name `cloud-run-secret`

**Impact:** Service account credentials properly injected

### 4. Health Check Timing

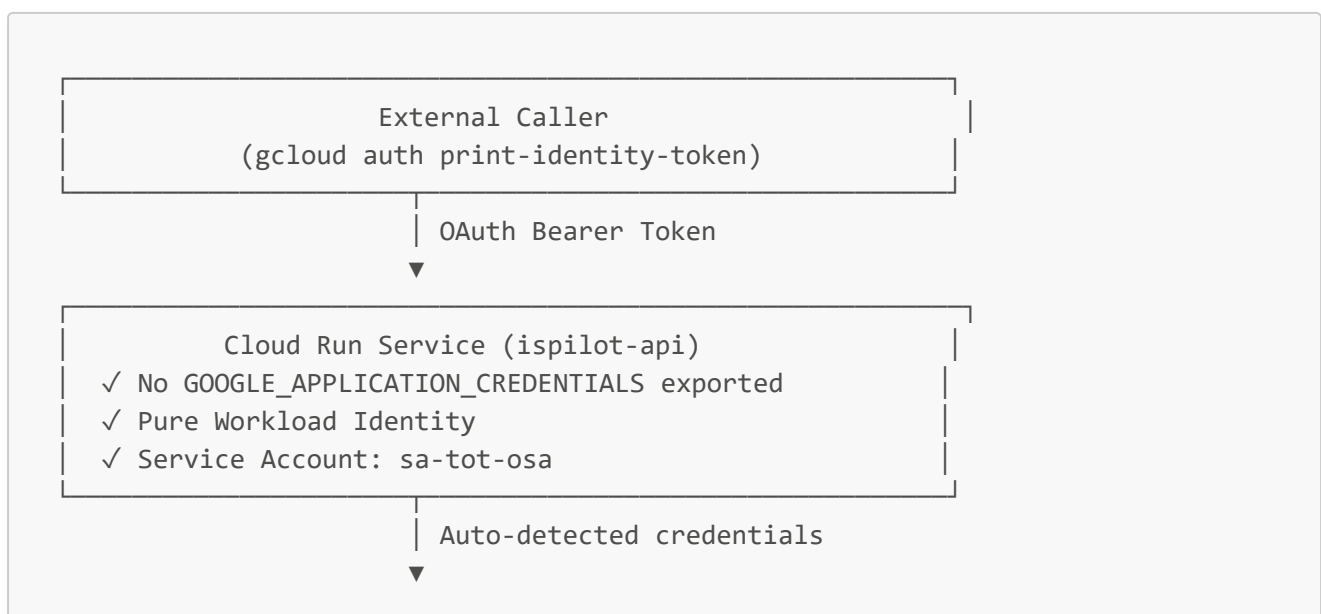
**Issue:** Health checks timing out during cold start

**Solution:** Increased wait time from 5s to 15s

**Impact:** Service fully warmed before health verification

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## Current Architecture



#### Google Cloud Services

- ✓ Vertex AI Reasoning Engine (5375474415045705728)
- ✓ Secret Manager (cloud-run-secret)
- ✓ Cloud Logging
- ✓ Firestore (optional, auto-detected)

## Authentication Flow

### Local Development

```
$ gcloud auth activate-service-account --key-file=/home/sduque/sa/key.json
$ AUTH_TOKEN=$(gcloud auth print-identity-token --audiences https://ispilot-api-46y2f3tyja-uc.a.run.app)
$ curl -H "Authorization: Bearer $AUTH_TOKEN" https://ispilot-api-46y2f3tyja-uc.a.run.app/chat
```

### Production (Cloud Run)

- Service account: `sa-tot-osa@corp-stro-salesinventory-prod.iam.gserviceaccount.com`
- Workload Identity binding automatically handles credentials
- No credentials file needed
- `google.auth.default()` automatically detects service account

## Test Results

### Successful Chat Query

```
$ curl -X POST "https://ispilot-api-46y2f3tyja-uc.a.run.app/chat" \
-H "Content-Type: application/json" \
-H "Authorization: Bearer $AUTH_TOKEN" \
-d '{"user_id": "test-user", "message": "Hello ISPilot"}'
```

### Response:

```
{
  "answer": "Hello! I am **IsPilot Analyst**, your assistant for analyzing
IsPilot performance KPIs...",
  "session_id": "8324292617089056768",
  "request_id": "2e9f0467-4b13-4b2f-b0c8-8dd74ffbb5a2",
  "timestamp": "2026-08-29T02:36:49.764889",
```

```
"status": "ok"
}
```

Status: ☒ SUCCESS

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## Deployment Process

### Automated Deployment (Recommended)

```
cd c:\Users\sduque\OneDrive - Falabella\Proyectos\2026\is\ispilot\intelligent-
shelf-agents\ispilot-api
bash deploy.sh
```

### What deploy.sh Does

1. Builds Docker image with multi-stage optimization
2. Configures Docker authentication to Artifact Registry
3. Pushes image to `us-central1-docker.pkg.dev/corp-stro-salesinventory-prod/ispilot-api/ispilot-api:latest`
4. Deploys to Cloud Run with:
  - Service account: `sa-tot-osa`
  - Memory: 1Gi
  - CPU: 2
  - Concurrency: 100
  - Timeout: 300s
5. Verifies deployment and health checks
6. Outputs service URL

### Deployment Time

- Build: ~2-3 minutes
- Push: ~1-2 minutes
- Deploy: ~3-5 minutes
- **Total:** ~10-15 minutes

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## Environment Configuration

### Production (Cloud Run)

```
GOOGLE_CLOUD_PROJECT      = corp-stro-salesinventory-prod
GOOGLE_CLOUD_LOCATION     = us-central1
VERTEX_PROJECT_ID         = corp-stro-salesinventory-prod
VERTEX_LOCATION           = us-central1
VERTEX_ENGINE_ID          = 5375474415045705728
```

```
FIRESTORE_COLLECTION      = user_sessions
SESSION_TIMEOUT_HOURS     = 8
```

## Local Development

```
GOOGLE_APPLICATION_CREDENTIALS = /home/sduque/sa/shelf-analyst-sa.json
GOOGLE_CLOUD_PROJECT           = corp-stro-salesinventory-prod
GOOGLE_CLOUD_LOCATION          = us-central1
VERTEX_PROJECT_ID              = corp-stro-salesinventory-prod
VERTEX_LOCATION                = us-central1
VERTEX_ENGINE_ID               = 5375474415045705728
ISPILOT_API_KEY                = test-key-for-local
SESSION_TIMEOUT_HOURS         = 8
```

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## Monitoring & Logs

### View Service Logs

```
gcloud run services logs read ispiilot-api \
  --project corp-stro-salesinventory-prod \
  --region us-central1 \
  --limit 50
```

### Check Service Status

```
gcloud run services describe ispiilot-api \
  --project corp-stro-salesinventory-prod \
  --region us-central1 \
  --format="table(status.url,status.revision,serviceAccountEmail)"
```

### Health Check

```
curl https://ispiilot-api-46y2f3tyja-uc.a.run.app/health
```

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## API Endpoints

### Health Check

```
GET /health
Response: {"status": "healthy", "timestamp": "..."}
```

## Chat Query

```
POST /chat
Headers: Authorization: Bearer <token>
        Content-Type: application/json
Body: {
  "user_id": "user-identifier",
  "message": "Natural language query",
  "session_id": "optional-session-id"
}
Response: {
  "answer": "Response from ISPilot",
  "session_id": "session-uuid",
  "request_id": "request-uuid",
  "timestamp": "ISO-8601",
  "status": "ok"
}
```

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## Troubleshooting

Issue: "Failed to authenticate with Vertex"

**Cause:** Cloud Run still has old deployment with hardcoded credentials

**Fix:** Delete old service and redeploy:

```
gcloud run services delete ispirot-api --quiet
bash deploy.sh
```

Issue: Health check timeout

**Cause:** Cold start taking longer than wait time

**Fix:** Increase **sleep** time in deploy.sh or check service logs

Issue: "Firestore API not enabled"

**Status:** Expected and handled gracefully

**Workaround:** Service falls back to in-memory sessions

**Enable when needed:**

```
gcloud services enable firestore.googleapis.com \
  --project corp-stro-salesinventory-prod
```

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## Next Steps

### Short-term (Optional)

1. Enable Firestore for persistent sessions
2. Set up Cloud Monitoring alerts
3. Configure Cloud Armor for DDoS protection

### Medium-term

1. Integrate with Microsoft Copilot Studio
2. Add Microsoft Teams connector
3. Implement rate limiting

### Long-term

1. Add more specialized agents
2. Implement recommendation engine
3. Build enterprise observability dashboard

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## Support

### Key Resources

- Deployment Guide: [ispilot-api/DEPLOYMENT.md](#)
- Architecture Docs: [docs/ISPilot-Enterprise-Architecture-And-Vertex-Agent-Engine-Guide.md](#)
- Platform Overview: [docs/ISPilot-Platform-Overview.md](#)

### Service Team

- Project: [corp-stro-salesinventory-prod](#)
- Region: [us-central1](#)
- Service Account: [sa-tot-osa@corp-stro-salesinventory-prod.iam.gserviceaccount.com](#)

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### Status Summary:

- **Authentication:** ☒ Workload Identity working
- **API:** ☒ Responding correctly
- **Vertex Integration:** ☒ Connected
- **Deployment:** ☒ Automated
- **Production Ready:** ☒ YES

**Last Updated:** August 28, 2026 @ 02:37 UTC

**Service Health:**  HEALTHY